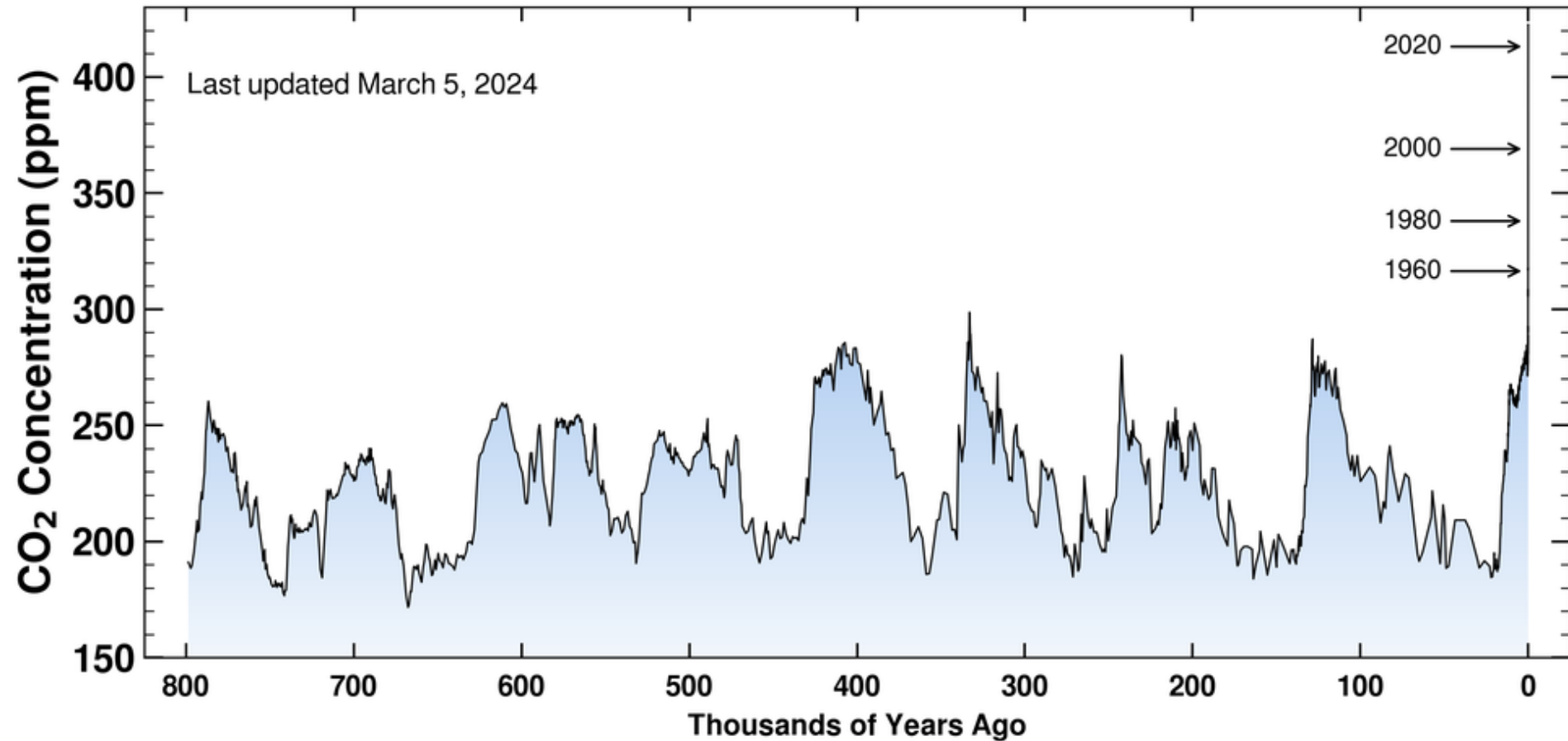


Lecture 11 – Time Series I



Today's Learning Outcomes

1. Know how to describe a periodic signal mathematically (i.e. period, amplitude, etc.)
2. Be able to explain how sampling interval and period of a signal can interact to create aliasing
3. Be able to explain the logic behind the turning points and phase length tests of signal
4. Be able to explain the logic behind the number of point increase and Kendall's Tau approaches to test for trends in a time series

Time Series

- A time series is a measurement of one or more variables over time
- The variables involved may be continuous (temperature, water head, etc.) or discrete (births, deaths, eruptions ,etc.)
 - But we almost always have time steps in which we have a time value, followed by a measurement of one or more variables

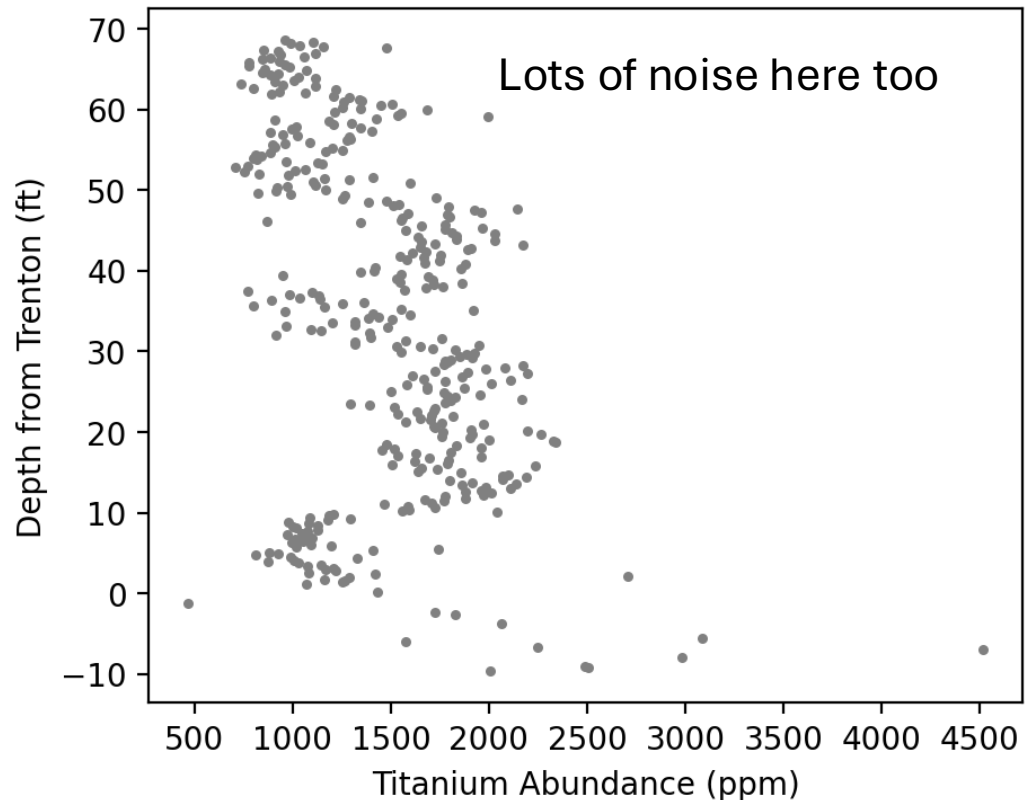
```
df = pd.read_csv("CoreDataNY10.csv")
plt.figure(figsize=[5,4],dpi=200)
plt.scatter(df['Ti'],df['DepthFromTrenton_ft'],s=5,c='gray')
plt.xlabel("Titanium Abundance (ppm)")
plt.ylabel("Depth from Trenton (ft)")
```

Core Data

Depth $\approx$ time

Common to observe both
directional and cyclic trends

Seasonal
Milankovitch Cycles
Basin-filling
Sea level cycles

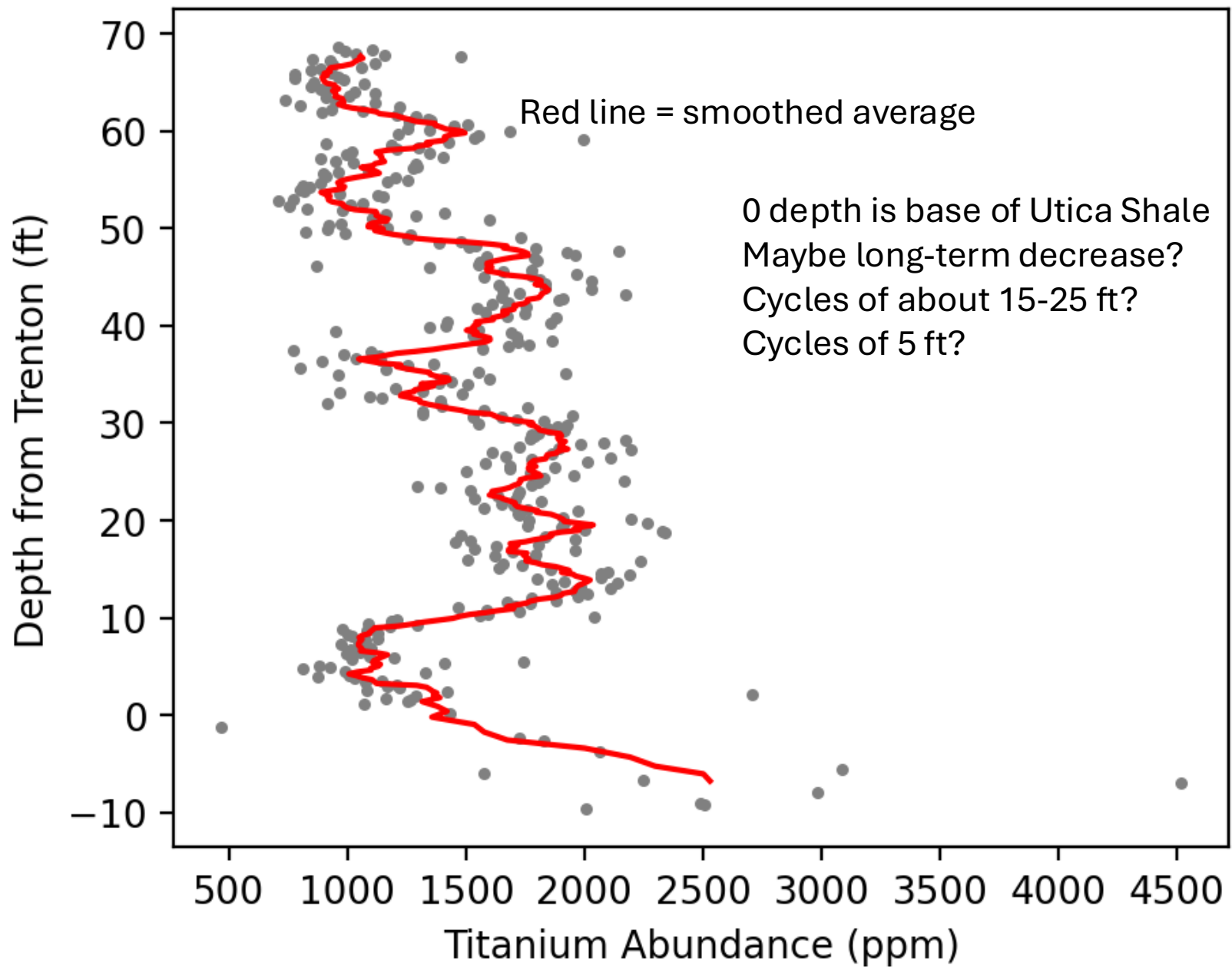


```
df = pd.read_csv("CoreDataNY10.csv")
plt.figure(figsize=[5,4],dpi=200)
plt.scatter(df['Ti'],df['DepthFromTrenton_ft'],s=5,c='gray')
plt.xlabel("Titanium Abundance (ppm)")
plt.ylabel("Depth from Trenton (ft)")
```

```
df_rolling_mean =
df[['DepthFromTrenton_ft','Ti']].rolling(window=10).mean().dropna()
plt.figure(figsize=[5,4],dpi=200)
plt.scatter(df['Ti'],df['DepthFromTrenton_ft'],s=5,c='gray')
plt.plot(df_rolling_mean['Ti'],df_rolling_mean['DepthFromTrenton_ft'
'],c='red')
plt.xlabel("Titanium Abundance (ppm)")
plt.ylabel("Depth from Trenton (ft)")
```

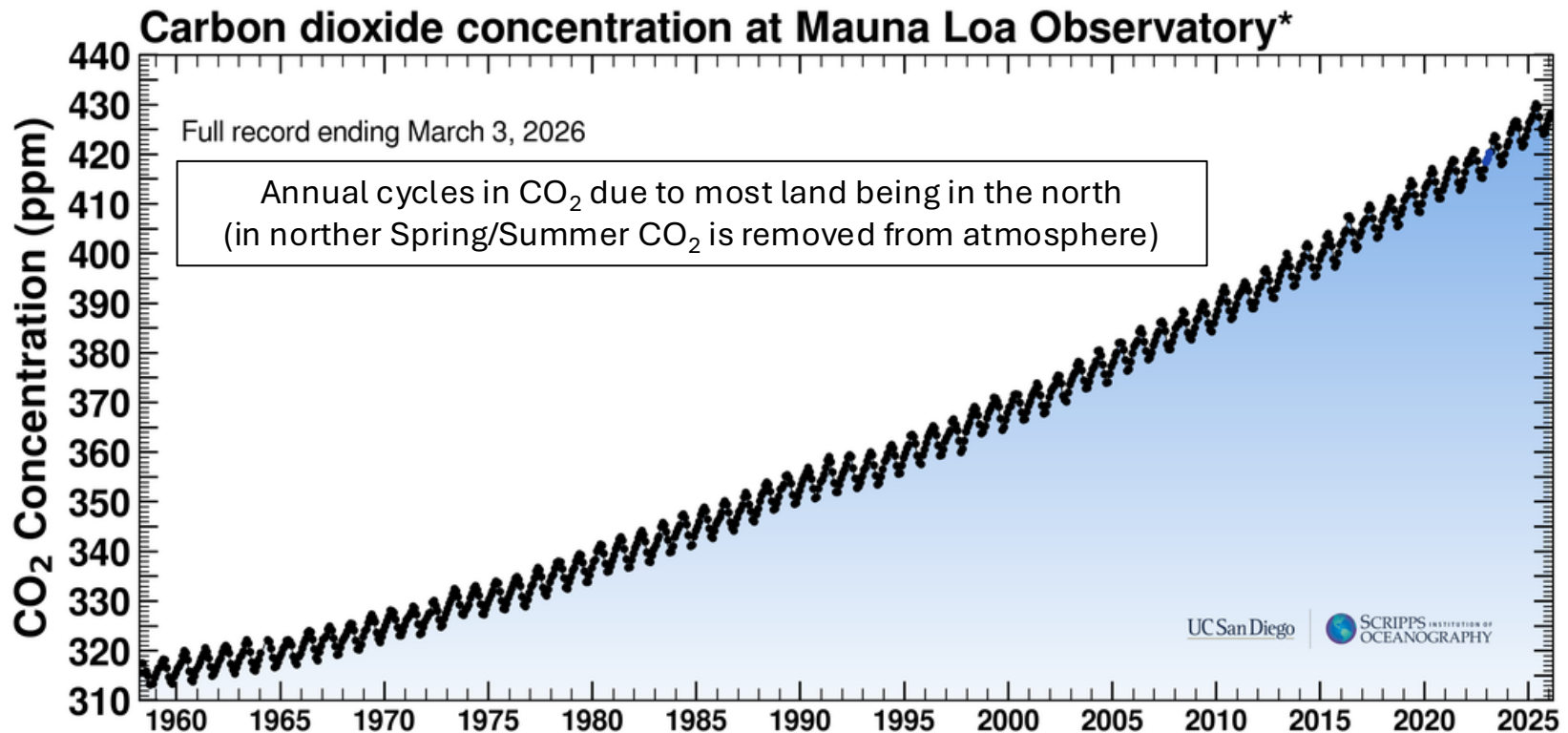
Use the `.rolling()` function from the pandas package to create a smoothed average to make the longer-term trends clearer

`window = 10` parameter means averaging over 10 points per step



Periodicity

- Time series often have repeating signals in them, meaning that some event repeats with a consistent time interval between events
 - Daily and yearly cycles are often obvious in physical systems

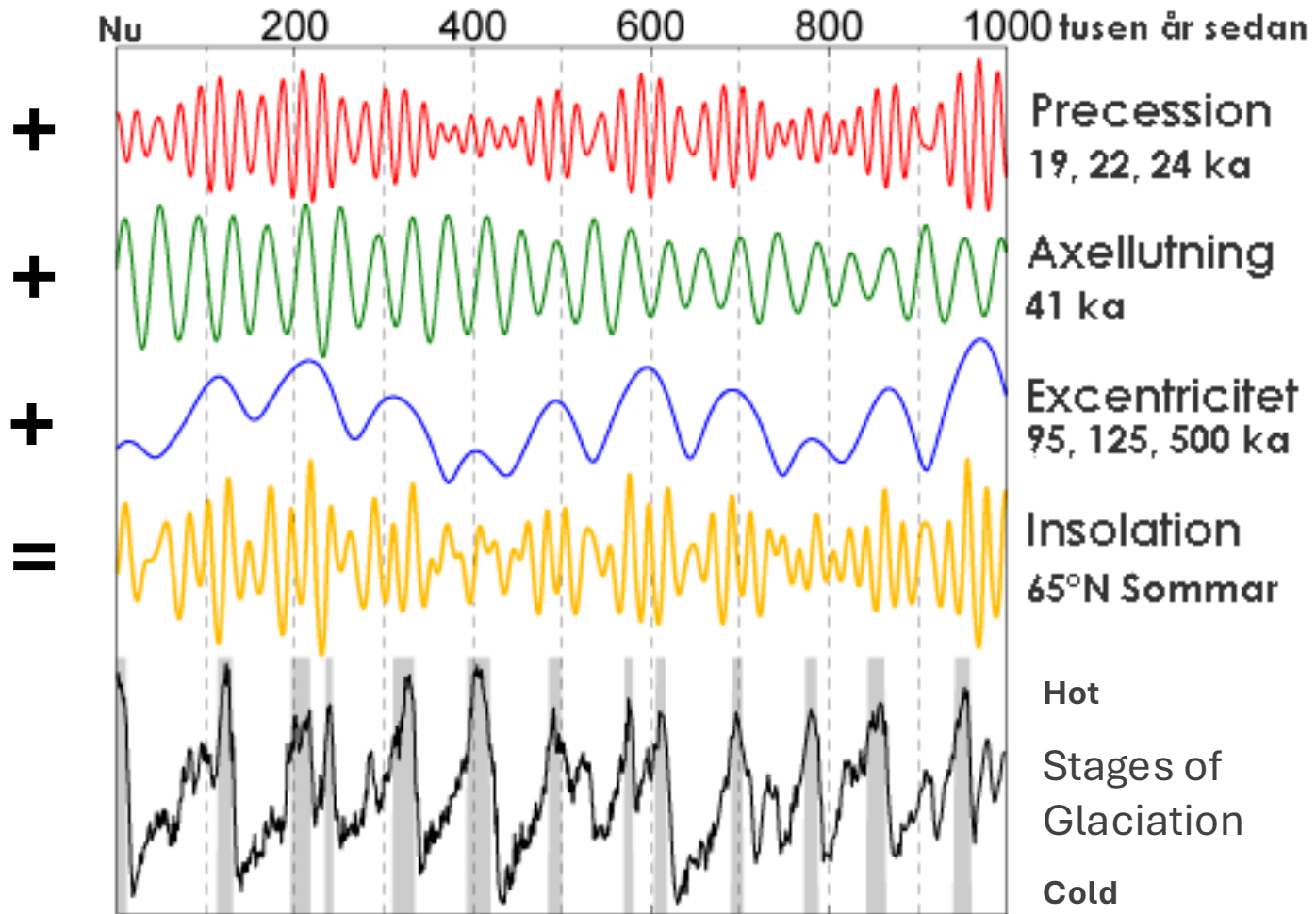


Periodicity

- Many methods in time series analysis are optimized to detect and analyze these types of periodic events
- Much early work in time series analysis came out of physics, which focused heavily on strictly periodic events
 - Newtonian physics provides clear cycles via interaction of gravitational pull of bodies on each other
- Some of those signals are the Milankovitch Cycles which effect climate (and all the things that go with that) on scales detectable in the sedimentary record
 - Can work backwards to create timescales from recognized cyclicity, known as **astrochronology**

Summed Periodic Signals

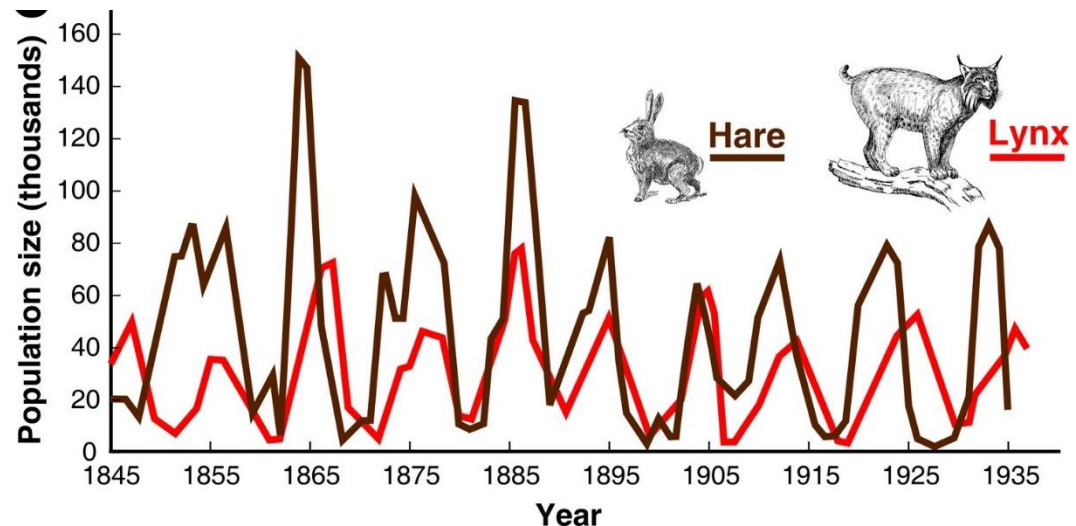
Common to have many periodic processes layered on each other



Aperiodic Repeating Events

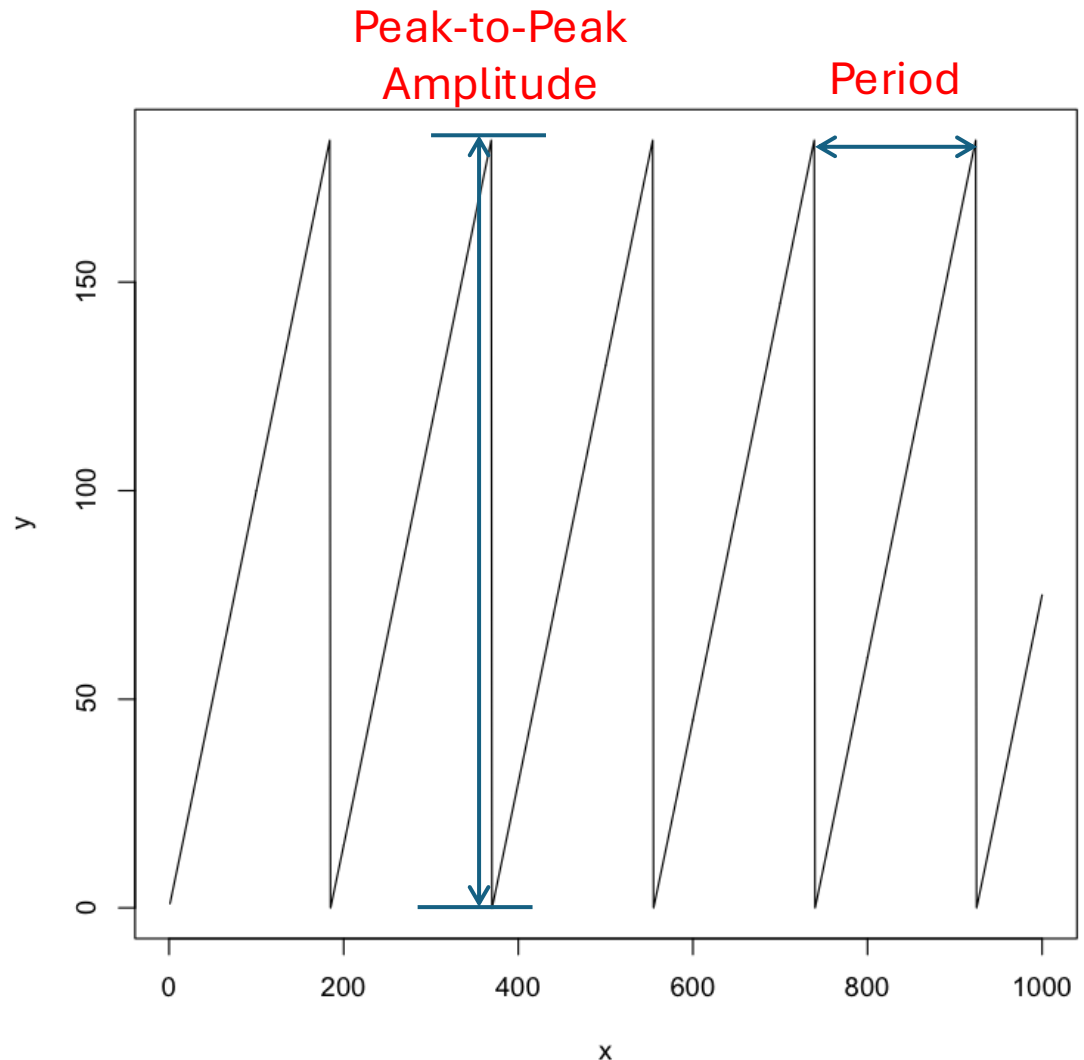
- Development of non-linear dynamics (aka chaos theory), makes is clear that there are also aperiodic repeating events
 - These are recurring events with a non-constant interval between events, but still deterministic
- Example: Predator-prey populations

Repeating maxima and minima in the populations, with a range of mean times between the maximum and minimum



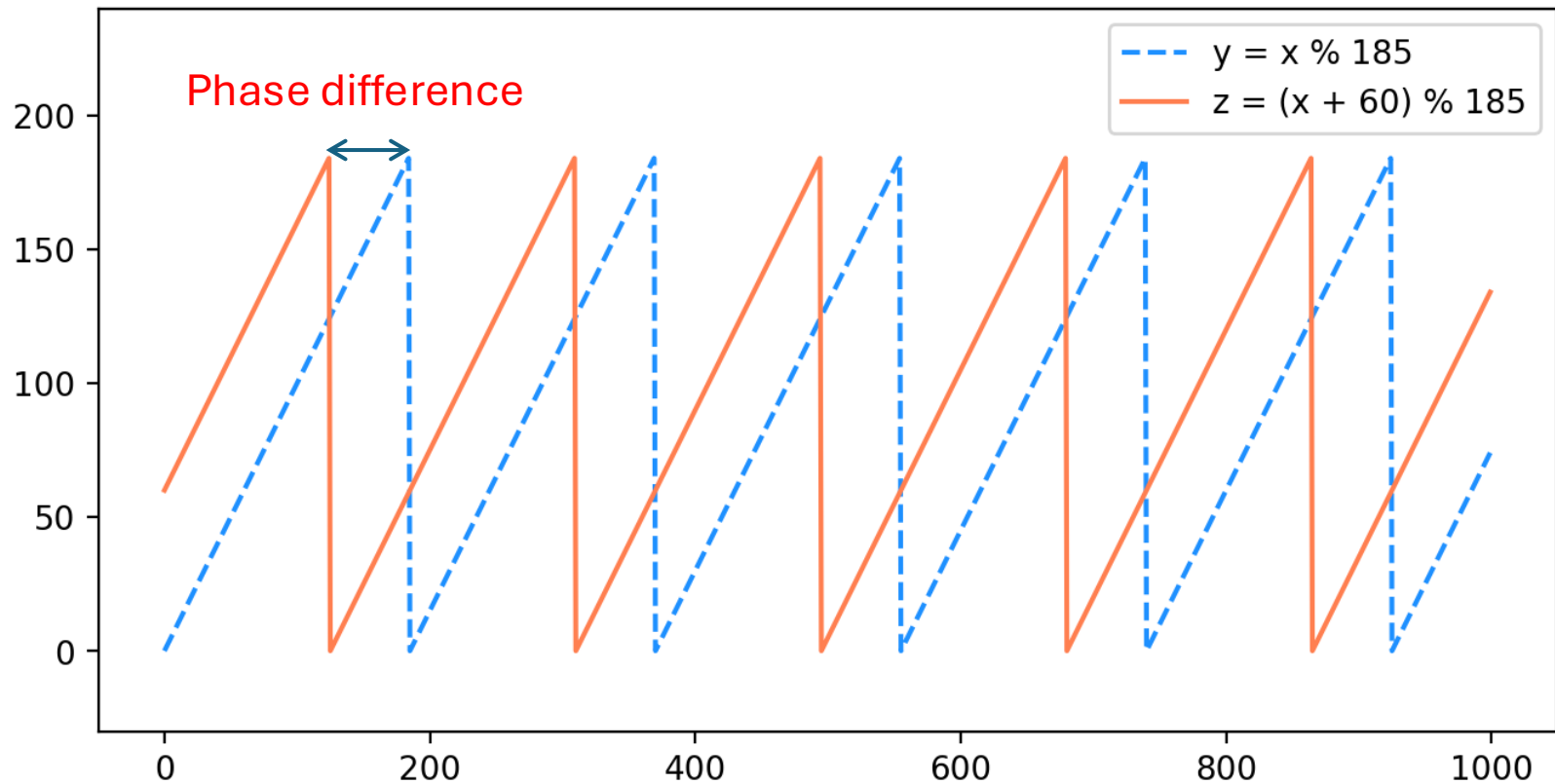
Describing Periodic Signals

- Period = time to complete 1 cycle
- Amplitude = $\frac{1}{2}$ peak to peak amplitude
 - i.e. peak to mean value distance
- Frequency = $1/\text{period}$



Describing Periodic Signals

- Sampling Rate or Sampling Interval = interval at which data is recorded
 - Ex. monthly temperature (interval is 1 month)
- Phase Shift = scaling factor where the only difference between two period signals is the timing of the peaks
- Phase differences often expressed as degrees (sine/cosine/tangent periodic signals)
 - $\text{Phase difference} = 360^\circ \times (\text{time between peaks}) / \text{period}$



Producing the “Sawtooth” wave used as an example here...

```
x = np.arange(0,1000)  
y = x % 185
```

I used the operator % to set the period to be 185

Adding 60 to x produced the phase shift

```
z = (x + 60) % 185
```

% is the remainder

$1 \% 185 = 1$

$2 \% 185 = 2$

.

.

$185 \% 185 = 0$

Length/Size of a Time Series

- We usually think of the size of a data set as being equal to n , the number of measured points
 - Generally, as n increases so does our information about a system
- Some odd problems when we start to talk about time series with periodicity
 - Increasing the number of points may or may not increase the amount of information we have

Length/Size of a Time Series

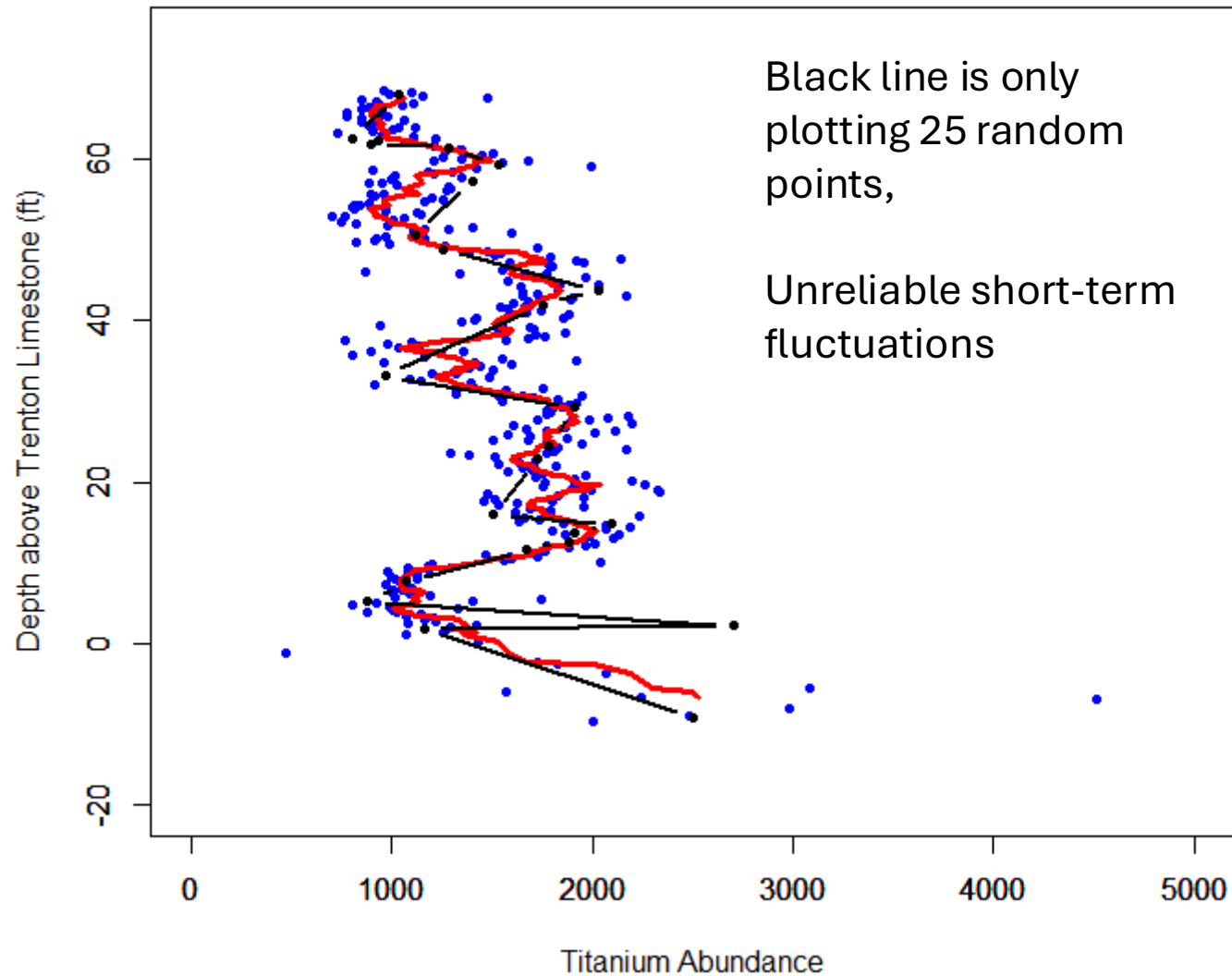
- Suppose we are interested in yearly temperature cycles, and have 1 measurement per week (at a consistent time, or better yet, averaged over the week) over 10 years, that's $n=520$
 - Suppose we decide to increase our measurements to once per day, so we have $n=3650$. How about once per hour, $n=85440$?
- Do we really learn much about yearly cycles of temperature by changing our sampling rate from weekly to daily to hourly?
 - Our n went up, but do we really know much more about yearly cycles?

Length/Size of a Time Series

- So the length of the time series needs to be measured as some sort of multiple of the period(s) of the events involved
 - If the events are aperiodic, we'll need to think in terms of the mean time between repetitions of events, or some other way of talking about a *characteristic time scale* of the phenomena we are interested in (i.e. distribution models)
- So how often do we need to sample???
 - Depends on what you want to be able to detect

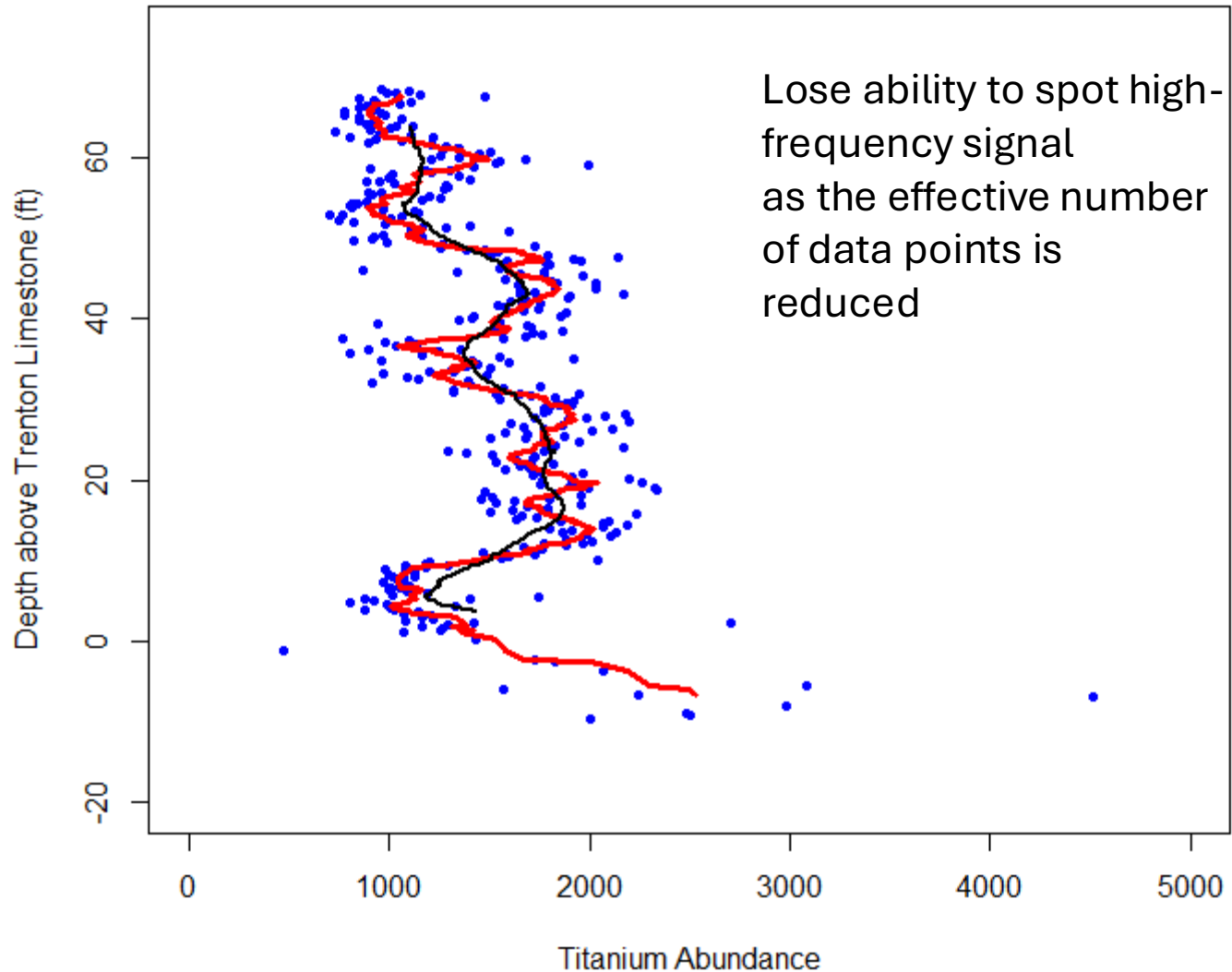
Nyquist Theorem

- To accurately record a signal, you must have at least two measurements during the period of the signal
- So the sampling period must be at least $\frac{1}{2}$ the period of the fastest event in your data, or the sampling frequency must be twice the frequency of the event
 - More is better, but no chance of detecting a real signal with less data



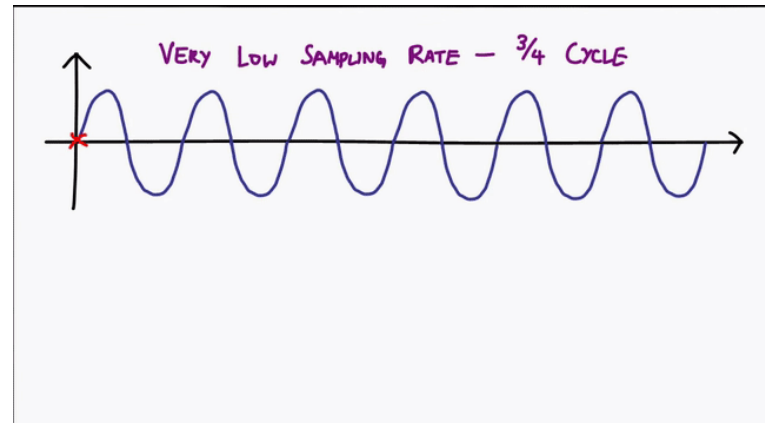
Red line = smoothed average (k=10)

Black line = smooth average (k=50)



Aliasing

- Worse, if you record values at lower frequency than the Nyquist frequency of the system, the high frequency signals will *alias* and mimic lower frequency signals!
- For this reason is it often wise to filter out very high frequency signals when you record data
 - Recording an average value over a time interval is probably better than a “snapshot” reading



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